Mohammed Sufiyan

Boston, MA (open to relocate) | LinkedIn | 857-263-1400
suff280602@gmail.com | GitHub | Google Scholar | Portfolio

EDUCATION

Boston University — **Boston, MA**
MS in Applied Data Analytics (GPA: 3.9) — *Sep 2025 - Dec 2026*

Presidency University — **Bengaluru, India**
B Tech in Computer Science and Engineering (Data Science) — *Sep 2020 - Jul 2024*

WORK EXPERIENCE

Power Tech — **Bengaluru, India**
Data Engineer — *Jun 2024 - May 2025*
- Cut processing time by **40%** with **25+** Python cleaning and transformation workflows across **17 datasets**.
- Improved reporting performance by **30%** through **40+** SQL queries and **5 Power BI dashboards** for daily operations.
- Processed **10GB+ daily** with AWS S3 and Lambda ETL and containerized **4 apps** with Docker, cutting deployment from **3 hours** to **under 30 minutes**.

Boston University | GitHub | Live Demo — **Boston, MA**
Graduate Research Assistant — *Jan 2026 - Aug 2026*
- Met **twice weekly** with professors to review sensor data, research protocols, deployment, dashboard changes, and BU grant proposals.
- Designed a **29 table PostgreSQL** schema for sensor, gateway, zone, reading, and alert data, built **Node.js** REST APIs for ingestion.
- Analyzed campus temperatures in **MATLAB** and deployed a **DigitalOcean** Leaflet dashboard for sensor trends, missing-data alerts, and CSV export.
- Presented the Urban Heat Island research at Boston University and **ACSP 2026** in Pittsburgh, Pennsylvania.

Boston University — **Boston, MA**
Graduate Teaching Assistant — *Sep 2026 - Present*
- Explain course concepts to **35 students** in Zoom sessions and grade technical assignments, providing clear feedback to support their learning.

PROJECTS

Data Quality and ETL Validation Framework | GitHub | Portfolio — *Oct 2026*
- Built a synthetic Brazilian ecommerce warehouse with PostgreSQL, Parquet, AWS S3, and Snowflake. Generated **65.4M records locally** and loaded **65,443 row sample** to Snowflake via an internal stage.
- Developed **10 dbt models**, one customer history snapshot, and **35 data tests**, all **46 dbt nodes passed** on the Snowflake sample.
- Built a Python regression suite with **pytest** and **97 passing tests**, then ran **59 Snowflake SQL checks** to confirm exact reconciliation across seven sample tables. Documented profiling results and test cases for each run.

Pipeline Health Monitor | GitHub — *Jan 2026*
- Validated **100,000 synthetic orders** in a local test with Apache Spark and orchestrated quality checks using Apache Airflow.
- Built dbt models and tests in Snowflake, tracked pipeline runs, and rejected records in MySQL and Streamlit, packaged with Docker Compose.

Card Fraud Detection and Alerting Platform | GitHub — *Jun 2026*
- Scored **10,000+** simulated card transactions in a local test with Python rules for amount, merchant category, country, channel, time, and device signals, returning a score and reason for each.
- Implemented a Redpanda streaming pipeline with PostgreSQL score storage and Redis alert caching, packaged with Docker Compose for API and dashboard access.

Formula 1 Pit Stop Analysis and Prediction | GitHub — *Nov 2025*
- Modeled **54 SQL Server tables** and a star schema, using Python ETL to clean **10,452 records** with **48 attributes** and add indexes and season partitioning.
- Compared Logistic Regression, SVM, and Random Forest. Random Forest led slow pit stop prediction with **93.8% test accuracy** and **0.875 F1 score**.

RESEARCH PAPERS

Pix2Pix GAN Lung Segmentation for Chest Radiographs | Published in IEEE | IEEE — *Sep 2024 - Apr 2025*
- Trained a Pix2Pix U-Net with TensorFlow and OpenCV, achieving **98.25% accuracy**, **98.05% Dice** on Montgomery, and **95.87% accuracy** on Shenzhen.

LLMs in Reinforcement Learning Comparative Analysis | Accepted in IEEE — *Nov 2025 - Present*
- Integrated Llama 2 7B and Sentence-BERT with PPO, raising mean DoorKey reward from **0.38 to 0.53** across 8 runs per method using policy guidance and reward shaping.

SKILLS

Data Quality & Testing: ETL Testing, Regression Testing, Test Automation, pytest, Source to Target Validation, Data Reconciliation, Data Profiling, Data Validation
Data Engineering & Databases: Python, Pandas, SQL, Snowflake, Snowflake SQL, dbt, PostgreSQL, SQL Server, MySQL, Redis, Apache Airflow, Apache Spark, Redpanda, ETL, Data Warehousing, Data Modeling, Star Schema, Dimensional Modeling
Cloud & Tools: AWS S3, AWS Lambda, Docker, Docker Compose, DigitalOcean, PM2, nginx, Power BI, Git
AI / ML & Backend: scikit-learn, TensorFlow, OpenCV, Llama, Sentence-BERT, PPO, MATLAB, Node.js, FastAPI, Streamlit, REST APIs